4 Magnetostatics

INTRODUCTION

- An electrostatic field is generally produced by static or stationary charges, whereas a static magnetic field (or magnetostatic field) is produced if the charges are moved with a constant velocity.
- A magnetostatic field is produced by a uniform current flow (direct current).
- The researcher Oersted has found the connection between electric fields and magnetic fields.
  Oersted demonstrated that when the charges are moving, they are surrounded by a magnetic field.
- The charges in motion are like a flow of charges that establish an electric current.
- Therefore, a current-carrying conductor is constantly surrounded by a magnetic field.
- If a time-invariant current flows, then a time-invariant magnetic field will be produced.
  The direct current (DC) is a static flow of current consequently, the conductor carrying a DC will produce a static magnetic field.
- Thus, the study of a static magnetic field existing in the given space, generated due to the flow of the direct current through a conductor is described as magnetostatics.
- Many concepts like induced e.m.f, motoring, transformer action, and so on are subject to magnetostatics.
  Therefore, the study of a static magnetic field (or commonly referred to as magnetostatics) has a significant part in electromagnetics.

Grey Matter Alert!

Magnetostatics is the sub-field of electromagnetics defining a static magnetic field.

- Before the detailed study of static magnetic fields, let us try to understand the fundamental properties of the magnetic field.
- To comprehend the properties of the magnetic field, let us consider a permanent magnet having two poles - North (N) and South (S). The area around a magnet where the effects of the magnet can be experienced is known as the magnetic field. The presence of such a field can be sensed with the help of a compass needle (by observing the deflections).
- Such a field is expressed by non-existent lines around the magnet, which are called attractive lines of power. These imaginary lines generally represent the magnetic field around the magnet and are often referred to as the magnetic force lines.
- The direction of the magnetic force lines outside the magnet is from the north-pole to the south-pole.

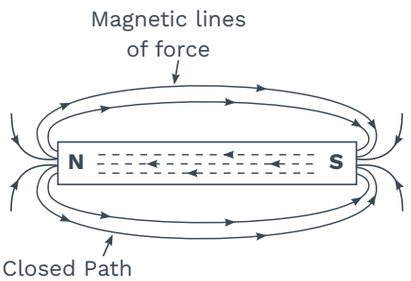


Fig. 4.1 Permanent Magnet

- The magnetic lines of force are also known as the magnetic lines of flux or magnetic flux lines.

Previous Years' Question

A solid iron cylinder is placed in a region containing a uniform magnetic field such that the cylinder axis is parallel to the magnetic field direction. The magnetic field lines inside the cylinder will:

a) Bend closer to the cylinder axis
b) Bend farther away from the axis
c) Remain uniform as before
d) Cease to exist inside the cylinder

- The electric flux lines originate from an isolated positive charge and deviate from terminating at infinity. And for a negative charge, the electric flux lines on a charge arising from infinity. However, in the case of magnetic flux, the poles exist in pairs only, and an isolated magnetic pole cannot exist.
- Thus, every magnetic flux line originating from the north pole should end at the south pole and follow the pathway from south to north inside the magnet.
- Therefore, magnetic flux lines exist in the form of a closed-loop.
- The magnetic flux line's existence is independent of the cause of the field generation, whether it is due to the conductor carrying direct current or due to a permanent magnet.

MAGNETIC FIELD DUE TO CURRENT-CARRYING CONDUCTOR

- As we know, a straight conductor carrying a direct-current produces a magnetic field around it by its length. The force lines are in the pattern of concentric circles in the planes at right angles to the conductor.

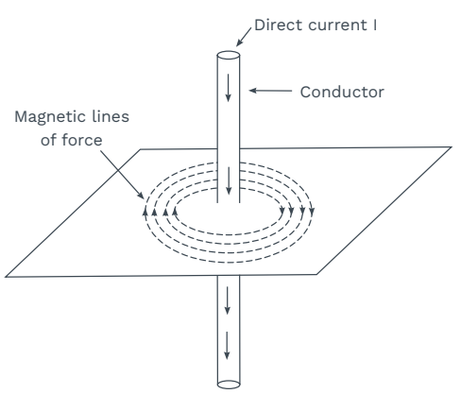


Fig. 4.2 Magnetic Field due to Conductor Carrying Direct Current

Rack Your Brain

The force experienced by an electro-magnetic wave in a conductor is:
a) Electrostatic force.
b) Magnetostatic force.
c) Electromotive force.
d) Lorentz force.

Previous Years' Question

A static magnetic field induces a current in a closed conducting loop.
a) True
b) False

- The direction of concentric circles is dependent on the direction of the current through the conductor.
- As the direction of current is fixed and the current is time-invariant, the magnetic lines of force are also constant, static, and time-invariant, producing a steady magnetic field around the conductor.
- The direction of the magnetic field around a direct current-carrying conductor can be determined by using the right-hand thumb rule.

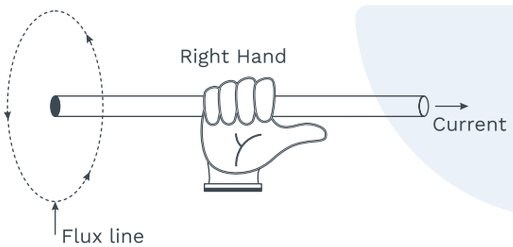


Fig. 4.3 Right-hand Thumb Rule

- It illustrates that, if we hold the current-carrying conductor in the right hand in such a manner that the thumb point in the direction of the current and parallel to the conductor. Therefore, the curled fingers will point in the direction of the magnetic lines of flux around it. Figure 4.3 demonstrates the right-hand thumb rule.
- The current-carrying conductor is represented by a small circle that is the top view of a straight conductor, whereas the current direction is indicated by a 'cross' or a 'dot'.
- The cross sign indicates that the current direction is into the plane or away from the observer.
- The dot sign shows that the current direction is coming out of the plane or towards the observer.

Rack Your Brain

If the surface area vector of a surface is perpendicular to the magnetic field, then the magnetic flux will be _____.

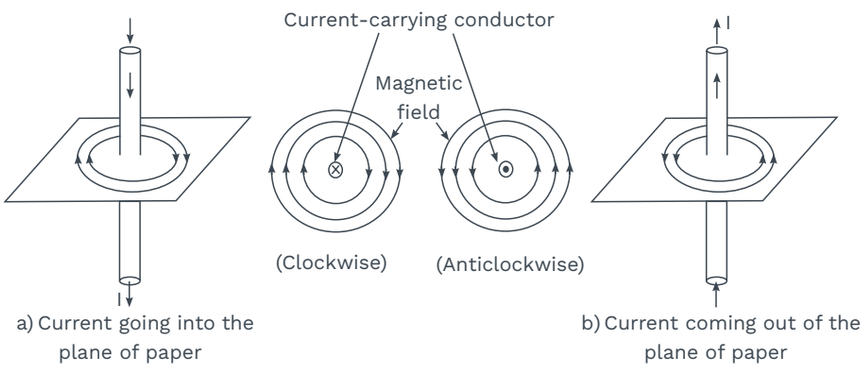


Fig. 4.4 Direction of Magnetic Flux Around a Conductor

- By employing the right-hand thumb rule, the direction of magnetic flux around a conductor is observed either clockwise or anticlockwise (figure 4.4).
- The total number of magnetic lines of force is described as magnetic flux and denoted as ϕ. The SI unit of magnetic flux is weber (Wb) and one weber means 10^8 lines of force.

Right-hand screw rule:

- When a right-handed corkscrew is rotated in such a way that its tip moves in the direction of the flow of current through the conductor, then the rotation of the head of the screw will provide the direction of the magnetic field, or if the right-handed screw is along the conductor carrying current with its axis parallel to the conductor, then the tip will point in the direction of the current flow.
- Also, the direction of the magnetic field can be determined by the direction in which the screw must be turned to advance in the direction of the current flow. Figure 4.5 demonstrates this rule.

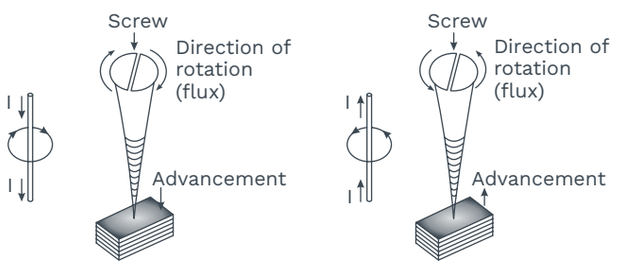


Fig. 4.5 Right-hand Screw Rule

Magnetic field intensity ($\vec{H}$):

- The quantitative measure of the strength of the magnetic field is termed as magnetic field intensity (or magnetic field strength).
- The magnetic field intensity at any point in the magnetic field is, referred to as the force endured by a unit north pole of one-weber strength when located at that point.
- The magnetic flux lines are evaluated in Weber (Wb), whereas the magnetic field intensity is measured in newtons/weber (N/Wb) or amperes per meter (A/m) or ampere-turns/meter (AT/m).
- The magnetic field intensity is symbolised by $\vec{H}$ and it is a vector quantity. The magnetic field intensity is like the electric field intensity ($\vec{E}$) in electrostatics.

Magnetic flux density ($\vec{B}$):

- The net magnetic lines of force or the magnetic flux crossing a unit area in a plane at right angles to the direction of flux is defined as magnetic flux density. It is denoted by $\vec{B}$ and is a vector quantity.
- It is measured in weber per square meter (Wb/m^2), which is also called Tesla (T).
- Magnetic flux density ($\vec{B}$) is like the electric flux density ($\vec{D}$) in electrostatics.

Relationship between $\vec{B}$ and $\vec{H}$:

- $\vec{E}$ and $\vec{D}$ in electrostatics are correlated to each other through permittivity $\in$ of the region.
- But in magnetostatics, the $\vec{B}$ and $\vec{H}$ are inter-linked with the property of the

Grey Matter Alert!

Permanent magnets are used for manufacturing because of:

- A broader hysteresis loops.
- High energy density.
- Positive susceptibility.
- A strong attraction to magnetic fields.

region in which the current-carrying conductor is placed. The property of the region is known as permeability, and it is symbolised by μ.

- It is a measure of the magnetisation obtained by a material (or a conductor) in response to the applied magnetic field.
- The permeability of the free space is denoted as μ_o (also called magnetic constant) and its value is $4\pi \times 10^{-7}$ H/m.
- As $\in$ is measured in F/m, while the permeability μ is measured in Henries per meter (H/m). For any other region, relative permeability is specified as $\mu = \mu_o\mu_r$
- The $\vec{B}$ and $\vec{H}$ are related as:

$$\vec{B} = \mu\vec{H} = \mu_0\mu_r\vec{H} \quad ...(4.1)$$

- For free space, $\vec{B} = \mu_0\vec{H}$
- For magnetic media, the permeability $\mu > 1$ while for non-magnetic materials, the permeability μ is unity.

BIOT-SAVART'S LAW

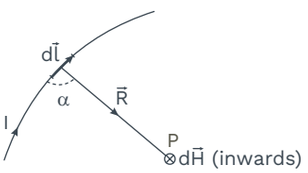


Fig. 4.6 Current Element Producing Magnetic Field Intensity.

Statement:

The magnetic field intensity $d\vec{H}$ produced at a point P due to differential current element $Id\vec{l}$ is generally:

- It is directly proportional to the product of the current (I) and the differential

Rack Your Brain

In an electromagnetic circuit, for developing a magnetic field:

i) The movement of the coil is needed.
ii) Energy need not be spent.
iii) Energy must be spent.
iv) Energy is not at all required.

Previous Years' Question

If two thin parallel wires are carrying the current along the same direction. The force experienced by one due to the other will be:

a) Zero
b) Perpendicular to the line and attractive
c) Perpendicular to the line and repulsive
d) Parallel to the lines

length (dl).

- It is also directly proportional to the sine of the angle between the element and the line joining point P to the element.
- It is inversely proportional to the square of the distance R between point P and the element.

$$dH \propto \frac{Idl \sin\alpha}{R^2}$$

$$dH = \frac{1}{4\pi}\frac{Idl \sin\alpha}{R^2} \quad ...(4.2)$$

- To represent magnetic field intensity $d\vec{H}$ in the vector form.

$$\left|d\vec{l} \times \hat{a}_R\right| = dl \times 1 \times \sin\alpha$$

Here, $|d\vec{l}|=dl$

$$|\hat{a}_R|=1=\frac{\vec{R}}{R}$$

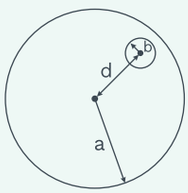
Therefore,

$$d\vec{H} = \frac{Id\vec{l} \times \hat{a}_R}{4\pi R^2} = \frac{Id\vec{l} \times \vec{R}}{4\pi R^3}(A/m) \quad ...(4.3)$$

- The direction of the magnetic field intensity can be determined by using the right-hand thumb rule.
- As the right-hand thumb rule says that thumb should be pointing in the direction of the current.
  The fingers of the right-hand encircling the given conductor will define the magnetic field direction.

Previous Years' Question

If an infinitely long uniform solid wire of radius 'a' carries a uniform DC current of density $\vec{J}$. A hole of radius 'b' (b < a) is now drilled along the length of the wire at a distance 'd' from the centre of the wire as shown below.



The magnetic field inside the hole is:

a) Uniform and depends only on 'd'.
b) Uniform and depends only on 'b'.
c) Uniform and depends on both 'b' and 'd'.
d) Non-uniform.

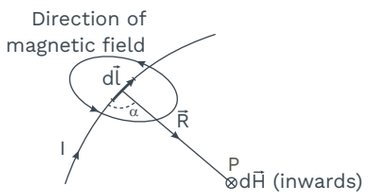


Fig. 4.7 Current Element Producing Magnetic Field Intensity

Case-I:

- At point P, if we encircle the fingers of the right-hand while keeping the thumb in the direction of the current.
  Then, it can be observed that the tip of the fingers is going inwards or inside the plane of point, i.e., the inward sign (denoted by cross ⊗).

Case-II:

- If the point P is at the location as shown in the figure below:

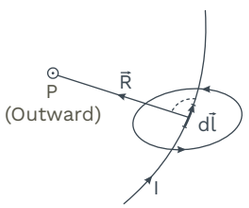


Fig. 4.8 Current Element Producing Magnetic Field Intensity

- It can be observed that, while using right-hand thumb rule and encircling the fingers and keeping the thumb in the direction of the current, then the tip of the fingers will come outward from the plane.
  Hence, it is marked as outwards (dot ⊙) in the figure.

Magnetic field due to finite-length straight current-carrying filament/conductor:

- Assume a straight current-carrying conductor of length AB (as shown in the figure below).

Grey Matter Alert!

Magnetostatics mainly depends on the inductance of the magnetic materials, which decides their effect on the influence of the magnetic field.

- We have to analyse the magnetic field at point P due to the finite length conductor.
- For that, let current (I) flows in the finite length conductor.
- Also, consider α_1 and α_2 angles.
  Here, the current is originates from point A.
  Thus, extend this point A.
  The axis at which the conductor is placed is referred to as the conductor axis.
- Now, join the current originating point (here, that point is A) and the point at which the magnetic field is to be determined.
- The angle between the extended line or conductor axis of points A and AP will be denoted as α_1.
  Similarly, join point B and point P.
  So, the angle between the conductor axis and BP will be denoted as α_2.
- Now, draw the perpendicular distance from the conductor to the point of interest (here it is P) or the distance between the conductor axis and the point of interest is denoted by ρ.

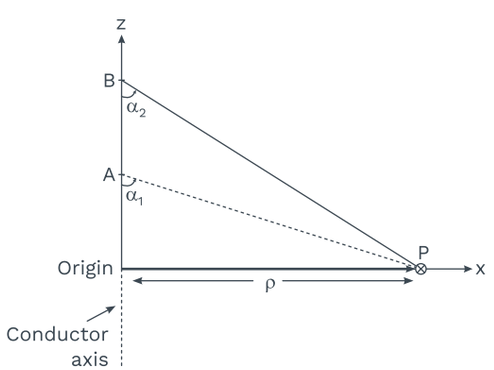


Fig. 4.9 Field at Point P due to a Straight Filamentary Conductor

Grey Matter Alert!

The two materials are differentiated by their permeability in the case of magnetic field and permittivity in the case of the electric field.

- Mathematically, the magnetic field at P due to finite straight length conductor is expressed by:

$$\vec{H} = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\hat{a}_\phi \quad ...(4.4)$$

- Here, the direction of the magnetic field can be determined by the right-hand thumb rule only.
  And the direction is inward, denoted by ⊗ cross or + y direction (inside the plane).

  $\hat{a}_l \times \hat{a}_\rho = \hat{a}_\phi$

  or $\hat{a}_z \times \hat{a}_x = \hat{a}_y$ (figure 4.10)

  Hence, positive y-direction.
  (∵ anticlockwise direction from x-direction because of right-hand thumb rule)
  $\hat{a}_l$ = Unit vector along the line current.
  $\hat{a}_\phi$ = Unit vector along the direction of the magnetic field.

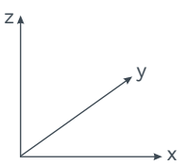


Fig. 4.10 Coordinate Axes

For the shapes other than straight finite length conductors, Biot-Savart's law will be applicable.

Case-I: For the semi-infinite length of the conductor.

- In this case, point A will be at the origin or we can say that conductor length initiates from origin.

Rack Your Brain

The following statement is correct/ incorrect?
"The tangential component of the magnetic field intensity is continuous at the boundary of separation of two media."

Rack Your Brain

Consider a bar magnet that is kept along the east-west direction on a horizontal surface in the earth's magnetic field. How many points will be there where the magnetic field is zero?

- As the length of the conductor is semi-infinite, thus, keeping a point A fixed at origin and point B at infinity (as shown in the figure below).

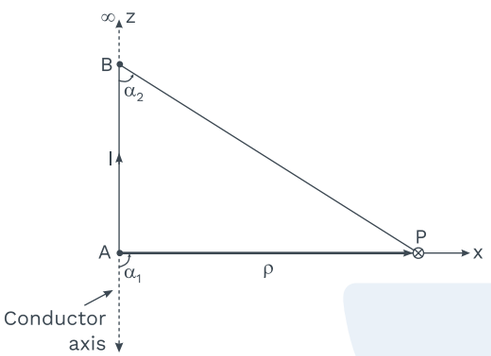


Fig. 4.11 Field at Point P due to a Straight Filamentary Conductor

- As point B heading towards infinity, the α_2 will become zero.

  Point B $\rightarrow \infty$; $\alpha_2 \rightarrow 0°$

  But the α_1 will be 90° in this case.

- Now, substituting values of α_1 and α_2 in the standard equations, as follows:

$$\vec{H} = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\hat{a}_\phi$$

As, α_1 = 90° and α_2 = 0°

$$\vec{H} = \frac{I}{4\pi\rho}(\cos 0° - \cos 90°)\hat{a}_\phi$$

$$\vec{H} = \frac{I}{4\pi\rho}\hat{a}_\phi \qquad ...(4.5)$$

Grey Matter Alert!

Magnetic lines of force are imaginary lines signifying the direction of the magnetic field such that the tangent at any point is in the direction of the field vector at that point.

Case-II: For the infinite length of the conductor.

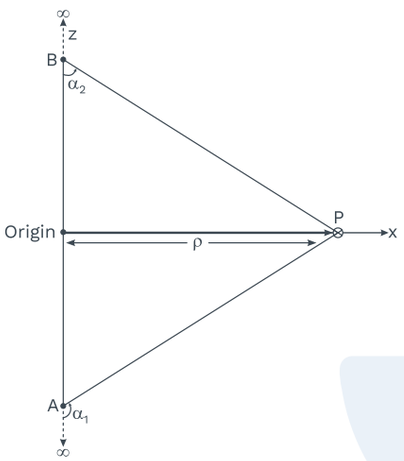


Fig. 4.12 Field at Point P due to a Straight Filamentary Conductor

- Here, α_1 will tends to 180° and α_2 tends to 0° for the infinite length of the conductor.
  Now, substituting values of α_1 = 180°, and α_2 = 0° in the equation (i), we get:

$$\vec{H} = \frac{I}{4\pi\rho}(\cos 0° - \cos 180°)\,\hat{a}_\phi$$

$$\vec{H} = \frac{I}{4\pi\rho}(2)\,\hat{a}_\phi = \frac{I}{2\pi\rho}\hat{a}_\phi \qquad ...(4.6)$$

Case III: If point P is near B and the B is at infinity (∞).

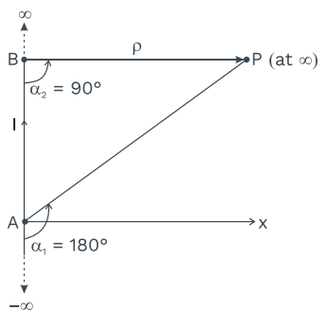
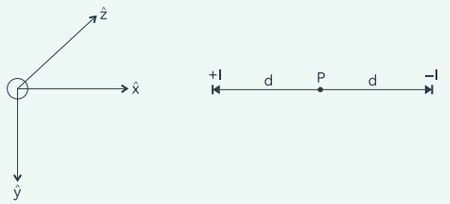


Fig. 4.13 Field at Point P due to a Straight Filamentary Conductor

Previous Years' Question

If two conductors are carrying forward and return current of +I and –I as shown in the figure, then the magnetic field intensity $\vec{H}$ at a point P is:



a) $\frac{I}{\pi d}\hat{y}$ **b)** $\frac{I}{\pi d}\hat{x}$

c) $-\frac{I}{2\pi d}\hat{y}$ **d)** $\frac{I}{2\pi d}\hat{x}$

Here, α_1 will tends to 180°, and α_2 tends to 90° for the infinite length of the conductor.

The magnetic field will be:

$$\therefore \vec{H} = \frac{I}{4\pi\rho}\hat{a}_\phi \qquad ...(4.7)$$

Magnetic field intensity due to infinite surface current/sheet:

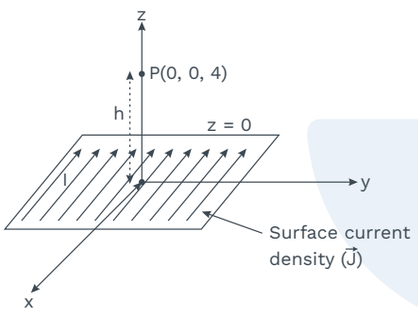


Fig. 4.14 Magnetic Field Intensity due to Infinite Surface Current

The magnetic field due to the infinite surface sheet is given by:

$$\vec{H} = \frac{1}{2}\left(\vec{J} \times \hat{a}_n\right) \qquad ...(4.8)$$

where $\vec{a}_n$ = unit normal vector directed from the current sheet to the point of interest (here it is P) to the surface.

Case: If the point of interest is in +y direction (like P), then $\hat{a}_n$ will be positive and $\vec{H}$ will be positive.

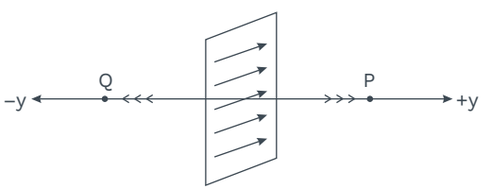
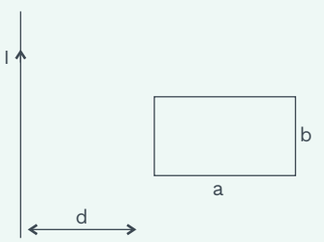


Fig. 4.15 Magnetic Field Intensity due to Infinite Surface Current

Previous Years' Question

The magnetic flux through a rectangular loop due to an infinitely long conductor of length 'a' and breadth 'b' as shown in the figure is:



If the loop and conductor are separated by distance d:

a) $\frac{\mu_0 Ib}{2\pi}\ln\left(\frac{d}{d+a}\right)$ Wb

b) $\frac{\mu_0 Ib}{2\pi}\ln\left(\frac{d+a}{d}\right)$ Wb

c) $\frac{\mu_0 Ib}{\pi}\ln\left(\frac{d}{d+a}\right)$ Wb

d) $\frac{\mu_0 Ib}{\pi}\ln\left(\frac{d+a}{d}\right)$ Wb

And if the point of interest is in a negative direction i.e. Q, then $\hat{a}_n$ will be negative. Hence, $\vec{H}$ will be negative.

- To reach the point of interest, sometimes we need to cover a distance and otherwise, we have to cover a perpendicular distance.
  Here, we have to analyse perpendicular direction (in $\hat{a}_n$) (here, distance does not matter)
- For a circular ring:

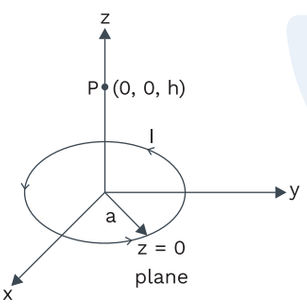


Fig. 4.16 Magnetic Field Intensity due to Circular Ring

$$\vec{H} = \frac{Ia^2}{2\left(a^2 + h^2\right)^{3/2}} \hat{a}_z \qquad ...(4.9)$$

The direction of $\vec{H}$:

If the current is flowing in the anticlockwise direction, and the circular ring is placed in a plane, then the direction of $\vec{H}$ will be in the perpendicular direction of the plane where this circular ring is placed.

- If a circular ring is placed in z = 0 plane, then the direction of $\vec{H}$ will be $+\hat{a}_z$ irrespective of the location of the point of interest P (above the ring or below the ring). Thus, the direction of magnetic field intensity will depend only on the direction of current flow.



Rack Your Brain

A charged particle goes unaccelerated in a region containing electric and magnetic fields, then:

a) Electric field ($\vec{E}$)must be parallel to $\vec{B}$.

b) Velocity ($\vec{v}$) must be perpendicular to the Electric field ($\vec{E}$).

c) Velocity ($\vec{v}$) must be parallel to magnetic flux density ($\vec{B}$).

d) Electric field ($\vec{E}$) must be equal to $\vec{v}$ $\vec{B}$.

Biot-ssavart law in a medium:

- If the conductor is placed in a medium, then the vector form of the magnetic field is expressed as follows-

$$d\vec{B} = \frac{\mu}{4\pi}\frac{I\left(d\vec{l}\times\vec{R}\right)}{|\vec{R}|^3} = \frac{\mu_0\mu_r}{4\pi}\frac{I\left(d\vec{l}\times\vec{R}\right)}{R^3} \quad ...(4.10)$$

  where μ_r = Relative permeability = μ/μ_0 and μ = absolute permeability of the medium.
- For air or vacuum, $\mu_r = 1$ (Non-magnetic material).

Biot-Savart's law in terms of current density ($\vec{J}$):

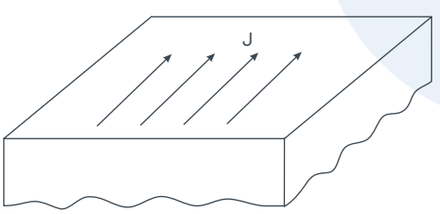


Fig. 4.17 Surface Current

- Let a surface carry a uniform current over its surface, as shown in the figure 4.17. Then the surface current density is denoted as $\vec{J}$ and it is measured in amperes per meter (A/m).
- As a result, for uniform current density, the current I in any area A is given by $I = \vec{J}A$,

$$\text{Since, } \vec{J} = \frac{I}{A} = \frac{Idl}{Adl} = \frac{Idl}{dV} \quad ...(4.11)$$

  And, $Id\vec{l} = \vec{J}dV$
- Therefore,

$$d\vec{B} = \frac{\mu}{4\pi}\frac{I\left(\vec{J}\times\vec{R}\right)}{R^3}dV \quad ... (4.12)$$

Previous Years' Question

The line integral of the vector potential around the boundary of a surface S represents:

a) Flux through the surface S.
b) Flux density in the surface S.
c) Magnetic density.
d) Current density.

Biot-Savart's law in terms of charge (q) and its velocity (v):

- A moving charge comprises current and thus, the corresponding magnetic field is given as equation (4.13)

$$d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{q\left(\vec{v} \times \vec{R}\right)}{R^3} \quad \ldots(4.13)$$

Since $Idl = \frac{q}{dt} dl = q\frac{dl}{dt} = qv$

$\therefore \quad Id\vec{l} = q\vec{v}$

- **Special conditions of Biot-Savart's law:**

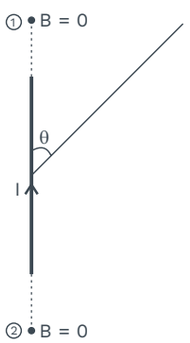


Fig. 4.18 Current Distribution

i) The magnetic field due to the current element can also be expressed as:

$$d\vec{B} = \frac{1}{4\pi \in_0 c^2} \frac{Id\vec{l} \times \vec{R}}{R^3} \quad \ldots(4.14)$$

where $c = \frac{1}{\sqrt{\mu_0 \in_0}}$ = Speed of light in vacuum.

ii) When θ = 0° or 180°, the magnetic field produced due to the current will be zero.

iii) When θ = 90°, i.e., R is perpendicular to dl, then the magnetic field will be $d\vec{B} = \frac{\mu_0 Idl}{\pi R^2}$ which is maximum.

Rack Your Brain

The torque (Nm) of a loop with magnetic moment 13.87 Am^2 and magnetic flux density 8.55 T is _____.

Magnetic field due to current in infinite straight wire:

- The direction of the magnetic field is the same due to all current elements. Then for the small current element:

$$B = \int dB = \frac{\mu_0}{4\pi}\int \frac{Idl\sin\theta}{r^2} \qquad ...(4.15)$$

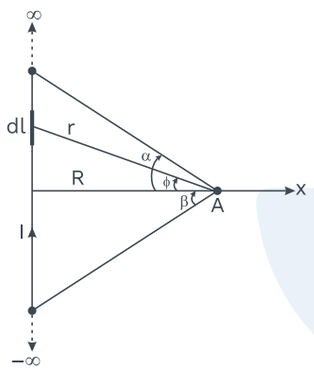


Fig. 4.19 Magnetic Field due to Current in Infinite Straight Wire

- We determine dl, r and ϕ in terms of perpendicular distance "R" (which is a constant for a given point) and angle "ϕ" in order to find B in terms of angle ϕ, here,

  $l = R\tan\phi$

  $dl = R\sec^2\phi\, d\phi$

  $r = R\sec\phi$

  where $\theta = \frac{\pi}{2} - \phi$

- By substituting the above equations in the integral, we get:

$$B = \frac{\mu_0}{4\pi}\int \frac{IR\sec^2\phi\, d\phi\, \sin\left(\frac{\pi}{2}-\phi\right)}{R^2\sec^2\phi}$$

$$B = \frac{\mu_0}{4\pi}\int \frac{I\cos\phi\, d\phi}{R} \qquad ... (4.16)$$

Grey Matter Alert!

When a magnetic specimen is subjected to a magnetising field, the magnetic moment created per unit volume is magnetisation that is represented by I and expressed as:

$I = \frac{M}{V}$

where M = Magnetic moment

and V = Volume.

- As I and R are constant, then after taking out of the integral equation 4.16 becomes as follows:

$$B = \frac{\mu_0 I}{4\pi R}\int \cos\phi \, d\phi \qquad ...(4.17)$$

$$B = \frac{\mu_0 I}{4\pi R}\int_{\alpha}^{\beta} \cos\phi \, d\phi$$

$$B = \frac{\mu_0 I}{4\pi R}(\sin\beta - \sin\alpha) \qquad ...(4.18)$$

- The angle (ϕ) is measured from the perpendicular line.
- The angle below the perpendicular line is considered as negative angle, whereas the angle above the perpendicular line is considered as positive angle.
  So, the sine of angle will be $\phi_1 = -\phi_2$ in above equation 4.18, we get:

$$B = \frac{\mu_0 I}{4\pi R}(\sin\alpha + \sin\beta) \quad ...(4.19)$$

Note: Here, the angles used in above equation are positive only.

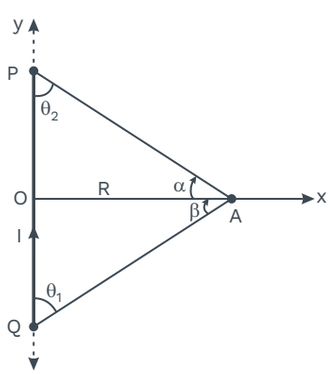


Fig. 4.20 Magnetic Field due to Current Finite Straight Wire

Previous Years' Question

Which of the following field equations shows that there are no free magnetic charges?

a) $\vec{H} = \frac{1}{\mu}\nabla \times \vec{A}$ **b)** $\vec{H} = \int \frac{Id\vec{l} \times \vec{R}}{4\pi R^3}$

c) $\nabla \cdot \vec{H} = 0$ **d)** $\nabla \times \vec{H} = \vec{J}$

Grey Matter Alert!

Magnetic polarisation is another name for magnetisation. The magnetic flux density in a material and in a vacuum are generally balanced in magnetic polarization.

- Also, the magnitude of the magnetic field depends on where the observation point 'A' is located with respect to the straight wire, which is reflected in the angle ϕ.
- The magnetic field corresponding to the current in a straight wire at a perpendicular distance "R" is commonly expressed in terms of angles between the straight wire and a line joining point of observation and end-points and is given as:

$$B = \frac{\mu_0 I}{4\pi R}\left[\sin\left(\frac{\pi}{2} - \theta_1\right) + \sin\left(\frac{\pi}{2} - \theta_2\right)\right]$$

$$B = \frac{\mu_0 I}{4\pi R}\left[\cos\theta_1 + \cos\theta_2\right] \quad \ldots (4.20)$$

The magnetic field at a point on the perpendicular bisector:

- The angles on both sides of the bisector are the same in this case.
  Here, $\alpha = \beta = \phi$

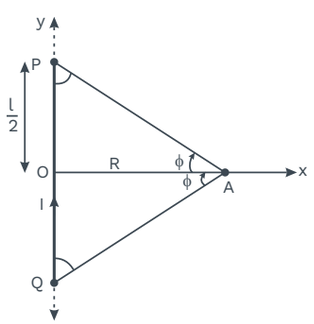


Fig. 4.21 Magnetic Field at a Point on Perpendicular Bisector

Thus, the magnetic field at a point on perpendicular bisector will be:

$$B = \frac{\mu_0 I}{4\pi R}\left[\sin\alpha + \sin\beta\right]$$

Rack Your Brain

Which of the statements below is FALSE?

i) Magnetic intensity is a vector quantity.

ii) Induced magnetisation is a method of magnetically modifying non-magnetic materials.

iii) The magnetic and magnetisation intensities are the same quantities.

iv) When a model of the earth's normal the magnetic field is removed the total magnetics intensity is calculated from the magnetometer.

$$B = \frac{\mu_0 I}{4\pi R}[2\sin\phi] \quad \{\because \phi_1 = -\phi_2\}$$

$$B = \frac{\mu_0 I \sin\phi}{2\pi R} \qquad ...(4.21)$$

- Let "l" be the length of the wire. Then,

$$\sin\phi = \frac{OP}{AP} = \frac{\frac{l}{2}}{\sqrt{\left(\frac{l}{2}\right)^2 + R^2}}$$

$$\sin\phi = \frac{l}{\sqrt{(l^2 + 4R^2)}} \qquad ...(4.22)$$

- From equation (4.21) and equation (4.22), we get-

$$B = \frac{\mu_0 I \sin\phi}{2\pi R} = \frac{\mu_0 I l}{2\pi R\sqrt{(l^2 + 4R^2)}} \qquad ...(4.23)$$

Note: The following points regarding the above equation:

i) For an infinitely long straight wire:

The magnetic field, due to the infinite can obtain wire by putting appropriate values of angles in the equation of the magnetic field due to the finite wire. Here, $\alpha = \beta = 90°$

$\sin\alpha + \sin\beta = \sin 90° + \sin 90° = 2$

$$B = \frac{\mu_0 I}{4\pi R}[\sin\alpha + \sin\beta]$$

$$B = \frac{\mu_0 I}{4\pi R}[\sin 90^\circ + \sin 90^\circ]$$

$$B = \frac{\mu_0}{2\pi}\frac{I}{R} \qquad ...(4.24)$$

ii) When a wire is semi-infinite (at the foot of long wire):

Grey Matter Alert!

A solenoid in electromagnetism is a coil with a length that is significantly greater than its diameter. A solenoid's the helical coil does not have to revolve around a straight axis to work.

Now, the wire is extended from an identified position to infinity in only one direction.

In this case, the angles involved are:

$\alpha = 0°$ and $\beta = 90°$

$$\therefore \quad B = \frac{\mu_0 I}{4\pi R}(\sin 0° + \sin 90°)$$

$$B = \frac{\mu_0 I}{4\pi R} \qquad ...(4.25)$$

iii) For the axial position of wire i.e., when point P lies on the axial position of the current-carrying conductor, the magnetic field at A will be zero i.e. B = 0

iv) The B-R graph for an infinitely long-straight wire is a rectangular hyperbola for $B \propto 1/R$.

$$B = \frac{\mu_0}{2\pi}\frac{I}{R}$$

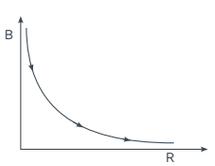


Fig. 4.22 The B-R graph

Rack Your Brain

The magnetic field inside a toroid is _____.

Magnetic field at the centre of a circular current-carrying coil:

- Here, a circular current-carrying coil is considered with radius 'a' and carrying current I.

  The magnetic field at the centre O, due to the current element dl is-

$$B = \frac{\mu_0 I}{2\pi a} \qquad ...(4.26)$$

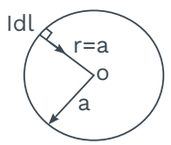


Fig. 4.23 Circular Current-carrying Coil

- The direction of the field in this case will be outward perpendicular to the plane of the paper.

The magnetic field due to an arc:

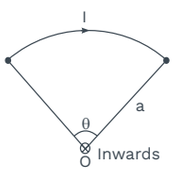


Fig. 4.24 Magnetic Field due to an Arc

- The magnetic field due to an arc of a circular current-carrying coil at the centre is given by,

$$B = \left(\frac{\theta}{2\pi}\right)\frac{\mu_0 I}{2a}$$

$$B = \frac{\mu_0}{4\pi}\left(\frac{I}{a}\right)\theta \qquad ...(4.27)$$

where θ is in radians.

Magnetic field at the axis of a circular current-carrying coil:

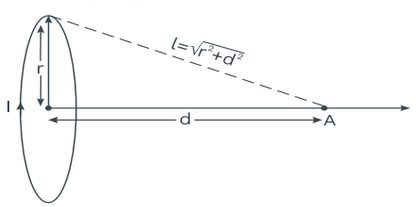


Fig. 4.25 Magnetic Field at the Axis of a Circular Current-Carrying Coil

- If we consider a coil with radius r and current I, the magnetic field on its axis at a distance d from its centre is given by:



Rack Your Brain

The magnetic moments of objects are generally measured with devices named as _____.

$$B = \frac{\mu_0}{4\pi}\frac{2\pi N I r^2}{l^3} \quad ...(4.28)$$

$$B = \frac{\mu_0}{4\pi}\frac{2\pi N I r^2}{\left(r^2 + d^2\right)^{3/2}} \quad ...(4.29)$$

where N = number of turns in the coil.

AMPERE'S LAW

Statement: The net current I enclosed by a the path is the same as the line integral of $\vec{H}$ around the closed path.

Alternatively, the total current is confined by a closed-loop is equal to the magnetic field intensity surrounding it.

Consider a current-carrying conductor as follows:

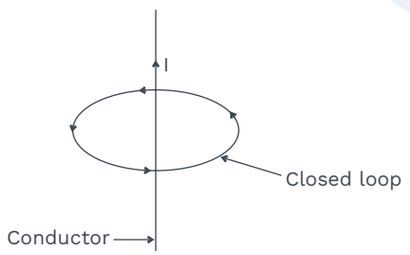


Fig. 4.26 Ampere's Law Applied to an Infinite Filamentary Line Current

Mathematically,

$$\oint_C \vec{H} \cdot d\vec{l} = I_{enc} \quad ...(4.30)$$

- Let $\vec{J}_C$ be the conduction current density (A/m^2), then

$$I = \int_S \vec{J}_C \cdot d\vec{S} \quad ...(4.31)$$

From equation (4.30) and (4.31), we get:

$$\oint_L \vec{H} \cdot d\vec{l} = \int_S \vec{J}_C \cdot d\vec{S} = \int_S \left(\nabla \times \vec{H}\right) \cdot d\vec{S} \quad ...(4.32)$$

Grey Matter Alert!

The magnetic energy density depends upon:
The magnetic field, due to the given the current distribution in the configuration.
And the permeability of the magnetic medium.

Previous Years' Question

The magnetic field intensity (in amperes/meter) at the centre of a circular coil of diameter 1 meter and carrying a current of 2 amperes is:

a) 8 **b)** 4
c) 3 **d)** 2

Equation (4.32) represents Maxwell's 3rd equation in integral form.

$$\nabla \times \vec{H} = \vec{J}_C \quad \text{...(4.33)}$$

Applications of Ampere's circuital law:

- The magnetic field associated with symmetric distributed current-carrying configurations can be evaluated by Ampere's circuital law.

I) Magnetic field due to a cylindrical wire:

- Consider cylindrical wire (r > R). Thus, to calculate the magnetic field outside at point A, we assume an amperian loop as shown below:

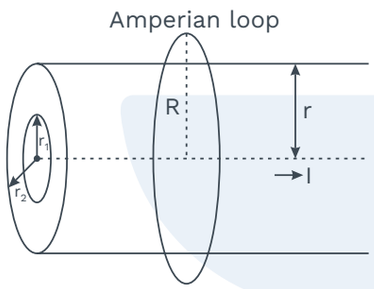


Fig. 4.27 Magnetic Field due to a Cylindrical Wire

- The magnetic field density is given as: $B = \frac{\mu_0 I}{2\pi r}$ ...(4.34)

II) Magnetic field due to a solenoid:

- A solenoid is considered, generally having a diameter of the coil shorter than its length.

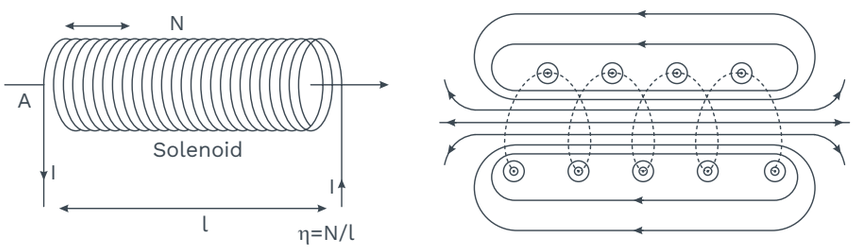


Fig. 4.28 Magnetic Field due to a Solenoid

i) For the finite length of solenoid:
If N = total number of turns, L= length of the solenoid then, n = number of turns per unit length = N/L.

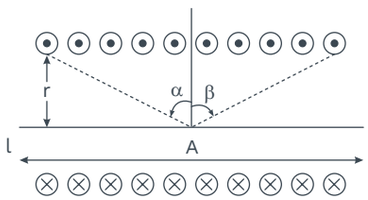


Fig. 4.29 Magnetic Field due to Finite Length of Solenoid

∴ The magnetic field inside the solenoid at point A is given by:

$$B = \frac{\mu_0 n I}{2} [\sin\alpha + \sin\beta] \quad ...(4.35)$$

ii) If the solenoid is of infinite length and the point is inside it then:

$$\alpha = \beta = 90°$$

And if the solenoid has infinite length and the point is near one end, then $\alpha = 0°$ and $\beta = 90°$

$$\therefore \quad B_{in} = \mu_0 n I$$

$$B_{end} = \frac{1}{2}\mu_0 n I$$

$$\text{or } B_{end} = \frac{1}{2}(B_{in}) \quad ...(4.36)$$

III) Magnetic field due to a toroid:

- A toroid is regarded as a ring-shaped closed solenoid.

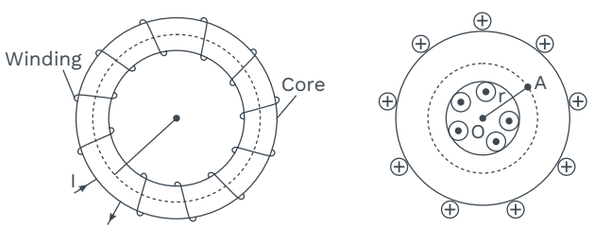


Fig. 4.30 Magnetic Field due to a Toroid

Rack Your Brain

If a 10V battery is used to power a circular circuit. When the battery is removed and put in the opposite direction, how does the magnetic field change?

a) It will become zero.
b) It will have the same magnitude but in the opposite direction.
c) It will have the same magnitude and direction.
d) None of the above.

- If a toroid has n turns per unit length. Then, the magnetic field at a point A in the figure is given as:

$$B = \frac{\mu_0 NI}{2\pi r} = \mu_0 n\, I \qquad \text{....(4.37)}$$

where $n = \frac{N}{2\pi r}$

IV) Magnetic field due to an infinitely large current-carrying sheet:

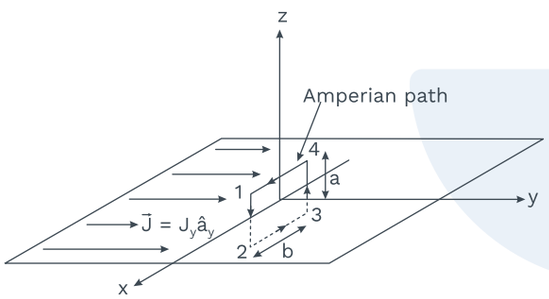


Fig. 4.31 Magnetic Field due to an Infinitely Large Current-carrying Sheet

- Let an infinite current sheet in the z = 0 plane.
- If the considered sheet has a uniform current density $\vec{J}$ (A/m) as shown in figure 4.31.
- The rectangular closed path 1-2-3-4-1 is studied using Ampere's law (Amperian path)

$$\oint \vec{H} \cdot d\vec{l} = I_{enc} = \vec{J}_y\, b \qquad \text{...(4.38)}$$

- Assume the infinite sheet is comprising of filaments.
- Due to a pair of filamentary currents, the resultant $d\vec{H}$ can be found either above or below the sheet and it has only x-components. In addition, $\vec{H}$ is negative on one side of the sheet and on the other hand, positive on the other side of the sheet.

Previous Years' Question

If the plane z = 0 carries surface current density $2\hat{a}_x$ A/m. The magnetic intensity H_y in the two regions: $-\infty < z < 0$ and $0 < z < \infty$ respectively will be in:

a) $\hat{a}_y$ and $-\hat{a}_y$ **b)** $-\hat{a}_y$ and $\hat{a}_y$
c) $\hat{a}_y$ and $-\hat{a}_x$ **d)** $-\hat{a}_x$ and $\hat{a}_x$

- Due to the infinite extent of the sheet, it is mainly regarded as filamentary pairs so that the characteristics of $(\vec{H})$ for a pair is the same for the infinite current-sheet:

$$\vec{H} = \begin{cases} H_0 \hat{a}_x & , \quad z > 0 \\ -H_0 \hat{a}_x & , \quad z < 0 \end{cases}$$

Where H_0 is to be determined as below.

- Evaluating the line integral of $\vec{H}$ in equation (4.38), along the closed path.

$$\oint \vec{H} \cdot d\vec{l} = \left[\int_1^2 + \int_2^3 + \int_3^4 + \int_4^1 \right] \vec{H} \cdot d\vec{l}$$

$$\oint \vec{H} \cdot d\vec{l} = 0(-a) + (-H_0)(-b) + 0(a) + H_0(b)$$

$$\oint \vec{H} \cdot d\vec{l} = 2H_0 b \qquad \ldots(4.39)$$

- From equations (4.38) and (4.39), we get:

$$H_0 = \frac{1}{2}(J_y)$$

$$\text{Now, } \vec{H} = \begin{cases} \frac{1}{2} J_y \hat{a}_x & , \quad z > 0 \\ -\frac{1}{2} J_y \hat{a}_x & , \quad z < 0 \end{cases}$$

- Thus, for an infinite sheet of current density $\vec{J}$ A/m.

$\vec{H} = \frac{1}{2} \vec{J} \times \hat{a}_n$ where $\hat{a}_n$ is a unit normal vector, which is directed from the current sheet to the point of interest.

- In terms of the magnetic field density,

$$\vec{B} = \frac{\mu_0 \vec{J}}{2} \qquad \ldots(4.40)$$

Magnetic field due to a Coaxial Cable:

- Consider a coaxial cable as shown in figure below-

Grey Matter Alert!

Magnetisability is the ratio of the magnetic moment to the magnetic flux density.

Rack Your Brain

What is the strongest part of a magnetic field called?

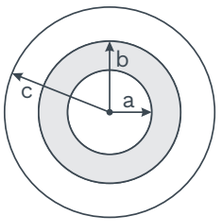


Fig. 4.32 Magnetic Field due to a Coaxial Cable

- In the first conductor, the radius is 'a' and the current flowing is I in the outward direction. The outer conductor has a radius 'b' and 't' is the thickness of the outer conductor along the outer conductor radius.
- Assuming the thickness of the outer conductor as 't' and radius as 'b'. So, we can say that c = b + t.
  And the current flowing inside the inner conductor is positive, and the current flowing inside the outer conductor is negative in nature.
- The outer conductor is in the form of a concentric-cylinder whose inner radius is b and the outer radius is c.
- This cable is placed along the z-axis. And the current I is uniformly distributed in the inner conductor.
- Let the space between the inner and outer conductor be filled with dielectric i.e. say air.
  Region 1: Magnetic field within the inner conductor 'r < a'. Consider a closed path having radius 'r < a'.

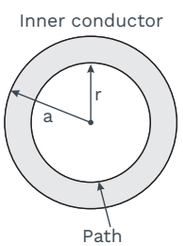


Fig. 4.33 For Radius r < a

Rack Your Brain

If a substance has permeability less than the permeability of free space, then it is known as _____.

Here, the cross-sectional area enclosed is πr^2. And the total current flowing is I through the area πa^2.
$\therefore$ The current enclosed by the closed path will be:

$$I' = \frac{\pi r^2}{\pi a^2} I = \frac{r^2}{a^2} I$$

The $\vec{H}$ is again only in $\hat{a}_\phi$ direction and depends only on r.

$$\therefore \vec{H} = H_\phi \hat{a}_\phi$$

- Considering $d\vec{l}$ in the $\hat{a}_\phi$ direction:

$$d\vec{l} = r\, d\phi\, \hat{a}_\phi$$

$$\therefore \vec{H} \cdot d\vec{l} = H_\phi\, \hat{a}_\phi \cdot r\, d\phi\, \hat{a}_\phi = H_\phi\, r\, d\phi$$

Thus, by Ampere's circuital law:

$$\oint \vec{H} \cdot d\vec{l} = \vec{I}'$$

$$\therefore \oint \vec{H} \cdot d\vec{l} = \frac{r^2}{a^2} I$$

$$\int_{\phi=0}^{2\pi} H_\phi r\, d\phi = \frac{r^2}{a^2} I$$

$$H_\phi\, r\,[2\pi - 0] = \frac{r^2}{a^2} I$$

$$\therefore H_\phi = \frac{r^2}{2\pi r a^2} I = \frac{r}{2\pi a^2} I$$

or $\vec{H} = \frac{Ir}{2\pi a^2} \hat{a}_\phi$ for $r < a$ (A/m) ...(4.41)

Rack Your Brain

How susceptibility defines the magnetic nature of the material?

Grey Matter Alert!

The substance will be paramagnetic if the susceptibility is positive whereas if the substance is diamagnetic, then susceptibility will be negative. Non-magnetic materials are classified as para- or diamagnetic if they do not have permanent magnetisation in the absence of an external magnetic field.

Region 2: This is the case of an infinitely long conductor along the z-axis where a < r < b. Consider a circular path that encloses the inner conductor carrying direct current.
Thus, $\vec{H}$ in this region, is:

$$H_\phi = \frac{I}{2\pi r}\hat{a}_\phi \text{ A/m}$$

$$B = \mu H = \frac{\mu_0 I}{2\pi r}\text{ Tesla} \qquad ...(4.42)$$

Region 3: Consider the closed path b < r < c as shown in figure 4.34.

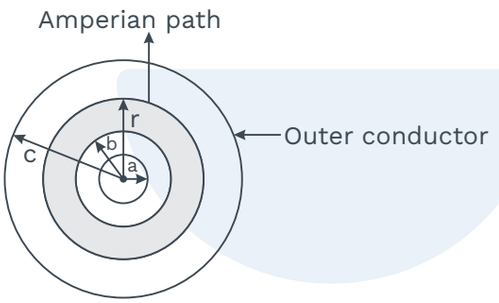


Fig. 4.34 Due to Closed Path

- The current enclosed by the closed path is:

$$I' = \frac{\pi\left(r^2 - b^2\right)}{\pi\left(c^2 - b^2\right)}(-I) = -\frac{\left(r^2 - b^2\right)}{\left(c^2 - b^2\right)}I$$

The total current (-I) is flowing through the cross-section $\pi(c^2\text{-}b^2)$ whereas the closed path encloses the cross-section $\pi(r^2\text{-}b^2)$.
Note: The closed path also encloses the inner conductor; hence the current I flows through it.

- The total current enclosed by the closed path is given by:

$$I_{enc} = I' + I''$$

$$I_{enc} = -\frac{\left(r^2 - b^2\right)}{\left(c^2 - b^2\right)}I + I$$

$$I_{enc} = I\left[1 - \frac{\left(r^2 - b^2\right)}{\left(c^2 - b^2\right)}\right] = I\left[\frac{c^2 - r^2}{c^2 - b^2}\right]$$

- According to Ampere's circuital law,

$$\oint \vec{H} \cdot d\vec{l} = I_{enc}$$

- Now $\vec{H}$ is again in ϕ-direction only and is a function of r only.

$$\vec{H} = H_\phi \hat{a}_\phi \text{ and } d\vec{l} = r d\phi \hat{a}_\phi$$

$$\vec{H} \cdot d\vec{l} = H_\phi \hat{a}_\phi \cdot r d\phi \hat{a}_\phi = H_\phi \, r d\phi$$

$$\therefore \int_{\phi=0}^{2\pi} H_\phi \, r d\phi = I_{enc} \cdot$$

$$H_\phi \, r[2\pi] = I\left[\frac{c^2 - r^2}{c^2 - b^2}\right]$$

$$\text{or } H_\phi = \frac{I}{2\pi r}\left[\frac{c^2 - r^2}{c^2 - b^2}\right]$$

$$\therefore \vec{H} = H_\phi \hat{a}_\phi = \frac{I}{2\pi r}\left[\frac{c^2 - r^2}{c^2 - b^2}\right] \hat{a}_\phi \text{ A/m} \quad ...(4.43)$$

Region 4: Outside the cable (r > c).

- Consider the closed path with (r > c) such that it encloses both the conductors with currents + I and – I.
  Hence, the total current enclosed is:

$$I_{enc} = +I - I = 0 \text{ A}$$

$$\oint \vec{H} \cdot d\vec{l} = 0$$

$$\vec{H} = 0 \text{ A/m}$$

∴ The magnetic field does not exist outside the cable. The variation of $\vec{H}$ against r is shown in figure 4.35.

Rack Your Brain

At the surface current, which of the magnetostatic boundary conditions are inaccurate?

a) The magnetic field's normal component is continuous.
b) The magnetic vector potential's normal component is continuous.
c) The magnetic vector potential's tangential component is continuous.
d) The magnetic vector potential's tangential component is not continuous.

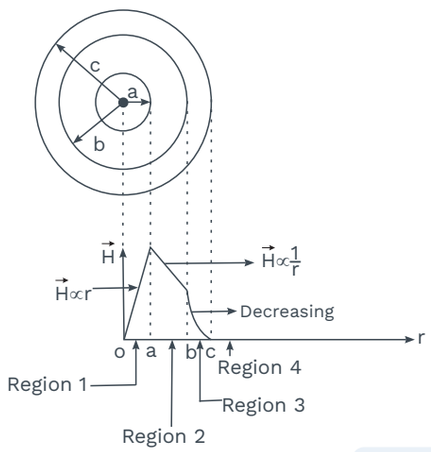


Fig. 4.35 Variation of Magnetic Field Intensity with Distance

Example 1:

Consider a filamentary current of $(10.2)\pi$ A flows in the negative z-direction. The magnetic field at (0,0,5) is:

a) 1.02 $\hat{a}_x$ A/m

b) 10 $\hat{a}_z$ A/m

c) 1.02 $\hat{a}_y$ A/m

d) 10.2 $\left(\hat{a}_x + \hat{a}_y\right)$ A/m

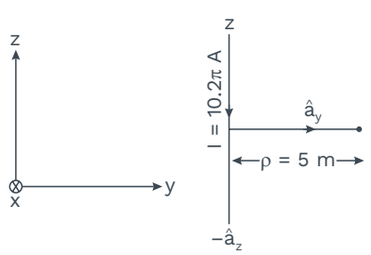
Rack Your Brain

If a square loop and a circular loop are made by a constant length current-carrying wire. Determine the ratio of magnitude field at the centre of the circular loop to the magnetic field at the centre of the square loop?

Sol:

Given that, I = $(10.2)\pi$ A

Perpendicular distance, ρ=5 units.



Thus, the magnetic field at (0,0,5) is given as:

$$\vec{H} = \frac{I}{2\pi\rho}\left(\hat{a}_\phi\right)$$

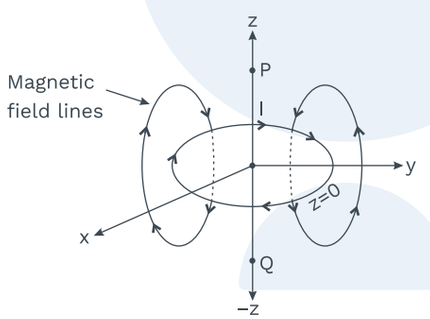
$$\vec{H} = \frac{(10.2)\pi}{2\pi(5)}\left(-\hat{a}_z \times \hat{a}_y\right)$$

$$= 1.02\ \hat{a}_x\ \text{A/m}$$

Hence, the correct option is A).

Example 2:
If the direction of current flow is clockwise in a circular ring placed at z=0 plane. What will be the direction of $\vec{H}$.

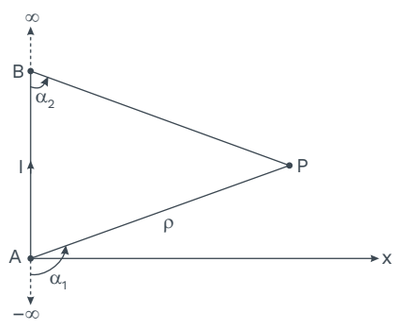
Sol:



$\vec{H}$ will create a north pole and direction will be only –z direction.
Hence, the direction of $\vec{H}$ will be:

$$\hat{a}_n = -\hat{a}_z$$

Example 3:
Calculate the magnetic field intensity $\vec{H}$ at point P (located as shown in the figure) due to the semi-infinite long current-carrying conductor.



Sol:
Here, $\alpha_2 = 0°$ as $\vec{B}$ is tending to infinity. But here, $\alpha_1 \neq 90°$.

So, the value of $\vec{H}$ can be determined accordingly by substituting the value of α_1 and α_2 in the equation. In the actual situation, α_1 will have values other than 90° and α_2 will be zero.

$$\vec{H} = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)\hat{a}_\phi$$

$$\vec{H} = \frac{I}{4\pi\rho}(\cos 0° - \cos\alpha_1)\hat{a}_\phi$$

$$\vec{H} = \frac{I}{4\pi\rho}(1 - \cos\alpha_1)\hat{a}_\phi \text{ A/m}$$

Example 4:
A loop with a radius of 0.23m and a current of 0.3A. Evaluate the magnitude of the magnetic field at the centre of the loop.

Sol:
Using the Biot-Savart law:

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{2\pi R^2 I}{(R^2 + \rho^2)^{3/2}}$$

Given, R is the radius of the loop = 0.23m, I is the current = 0.3A, and ρ is the distance from the centre of the loop = 0.

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$B = \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{2\pi(0.23)^2(0.3)}{[(0.23)^2 + 0^2]^{3/2}}$$

$$B = 8.19 \times 10^{-7} \text{ T}$$

Example 5:
Consider a square conducting loop of side 2a lies in z = 0 plane carries a current I in a counterclockwise direction. The $\vec{H}$ at the centre of the loop is _____.

Previous Years' Question

The magnetic vector potential $\vec{A}$ due to any current element Idl at any distance r will be:

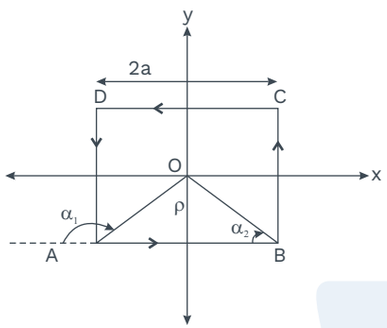
a) $\frac{Idl}{4\pi r}$ **b)** $\frac{Idl}{4\pi r^2}$

c) $\frac{\mu Idl}{4\pi r^2}$ **d)** $\frac{\mu Idl}{4\pi r}$

Sol:

Consider z = 0 plane i.e. x-y plane. Join line AO since it makes $\alpha_1 = 135°$.

Here, $\alpha_1 = 135°$ and $\alpha_2 = 45°$



As, $\vec{H} = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)$

$\vec{H}_1$ due to AB conductor will be:

$$\tan 45° = \frac{\rho}{a}$$

Thus, $\rho = a$

$$\therefore \vec{H}_1 = \frac{I}{4\pi a}(\cos 45° - \cos 135°)$$

Here, $\hat{a}_\phi = \hat{a}_l \times \hat{a}_\rho = \hat{a}_x \times \hat{a}_y = \hat{a}_z$

$$\therefore \vec{H}_1 = \frac{I}{4\pi a}\left[\frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right)\right]\hat{a}_z$$

$$\vec{H}_1 = \frac{I}{4\pi a}\left(\frac{2}{\sqrt{2}}\right)\hat{a}_z$$

$$\vec{H}_1 = \frac{I}{2\sqrt{2}\,\pi a}\hat{a}_z$$

Since, there are 4 sides, the total $\vec{H}_{net}$ will be:

Previous Years' Question

A coil of 300 turns is wound on a non magnetic core having a mean circumference of 300 mm and a cross-sectional area of 300 mm^2. The inductance of the coil corresponding to a magnetising current of 3 A will be (Given that $\mu_0 = 4\pi \times 10^{-7}$ H/m)

a) 37.6 μH
b) 113.04 μH
c) 37.68 mH
d) 113.04 mH

Rack Your Brain

Why is magnetic potential $\vec{A}$ a vector while the energy is scalar?

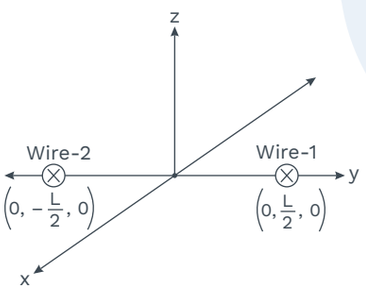
$$\vec{H}_{net} = 4\vec{H}_1$$

$$\vec{H}_{net} = \frac{4I}{2\sqrt{2}\,\pi a}\hat{a}_z$$

$$\therefore \vec{H}_{net} = \frac{\sqrt{2}\,I}{\pi a}\hat{a}_z \text{ A/m}$$

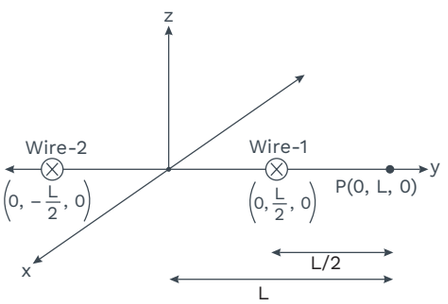
Example 6:

If a steady current(I) is flowing in the x-direction across both of the two infinitely long wires at y = ± L/2 as shown in figure below. The $\vec{H}$ at (0, L, 0) will be:



Sol:

Given,



For infinite long current-carrying conductor:

$$\vec{H} = \frac{I}{2\pi\rho}\hat{a}_\phi$$

$$\vec{H} = \frac{I}{2\pi\left(\frac{L}{2}\right)}\hat{a}_z + \frac{I}{2\pi\left(\frac{3L}{2}\right)}\left(-\hat{a}_z\right)$$

Previous Years' Question

Consider a current-carrying wire current 2A which is placed along the x and y-axis. Find $\vec{H}$ at point (5,5,0).

a) $\frac{1}{10\pi}(2+\sqrt{2})(-\hat{a}_z)$

b) $\frac{1}{10\pi}(2-\sqrt{2})(-\hat{a}_z)$

c) $\frac{1}{10\pi}(2-\sqrt{2})(\hat{a}_z)$

d) $\frac{1}{10\pi}(\sqrt{2}-2)(-\hat{a}_z)$

Here, $\hat{a}_\phi = -\hat{a}_x \times \hat{a}_y = -\hat{a}_z$

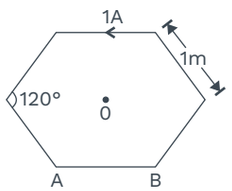
$$\vec{H}_{net} = -\frac{I}{\pi L}\hat{a}_z - \frac{I}{3\pi L}\hat{a}_z$$

$$\vec{H}_{net} = -\left[\frac{3I\hat{a}_z + I\hat{a}_z}{3\pi L}\right]$$

$$\therefore \vec{H}_{net} = -\frac{4I}{3\pi L}\hat{a}_z \text{ A/m}$$

Example 7:

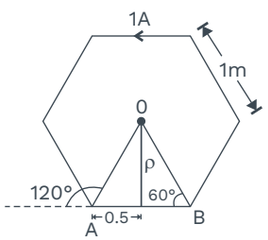
The magnitude of $\vec{H}$ at the centre of the loop of the wire-wound as a regular hexagon having side length 1m with 1A current flowing through it will be _____.



Sol:

Since, the given conductor is of finite length.

$\therefore$ Extending the side length (AB) of hexagon.



Here, $\alpha_2 = 60°$

and $\tan 60° = \frac{\rho}{0.5}$

$$\rho = 0.5\sqrt{3}$$

Previous Years' Question

The energy stored in the magnetic field of solenoid 30 cm long and 3 cm diameter wound the 1000 turns of wire carrying a current of 10 A is:

a) 0.015 J
b) 0.15 J
c) 0.5 J
d) 1.15 J

Rack Your Brain

If the magnetic potential is given as $\vec{A} = \int \vec{B} \times d\vec{l}$. Then, find $\vec{B}$?

$$\therefore \vec{H}_1 = \frac{I}{4\pi\rho}(\cos\alpha_2 - \cos\alpha_1)$$

$$\vec{H}_1 = \frac{1}{4\pi\left(0.5\sqrt{3}\right)}(\cos 60° - \cos 120°)$$

(since, α_1 = 120° from the figure)

$$\vec{H}_1 = \frac{1}{2\pi\sqrt{3}} \left(\frac{1}{2} + \frac{1}{2}\right)$$

$$\vec{H}_1 = \frac{1}{2\pi\sqrt{3}}$$

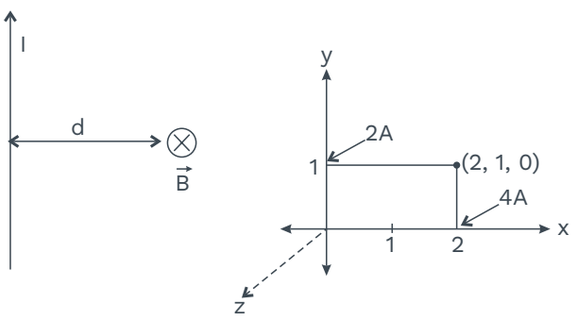
Thus, the total magnetic field intensity:

$$\vec{H}_{net} = 6\vec{H}_1$$

$$\vec{H}_{net} = 6 \times \frac{1}{2\pi\sqrt{3}} = 0.55 \text{ A/m}$$

Example 8:
Consider the magnitude of magnetic flux $(\vec{B})$ at a point with a normal distance 'd' meter from an infinitely extended-wire carrying 1A of current is $\frac{\mu_0 I}{2\pi d}$.

If this infinitely extended-wire is located along the axis carrying 4A current in the positive x-direction. Another infinitely extended-wire is placed along the y-axis and it carries a 2A current in the positive y-direction. Assume $\hat{i}, \hat{j}$ and $\hat{k}$ to be the unit vectors along x, y, and z directions, respectively.



Assume right-handed coordinate system, then the magnetic field intensity $\vec{H}$ at coordinate (2, 1, 0) will be:

a) $\frac{3}{2\pi}\hat{k}$ Wb/m^2 **b)** $\frac{4}{3\pi}\hat{i}$ Wb/m^2

c) $\frac{3}{2\pi}\hat{k}$ A/m **d)** 0 A/m

Sol:

The magnetic field intensity $\vec{H}$ along the x-direction:

$$\vec{H}_x = \frac{I}{2\pi\rho}$$

Here, $\rho = 1$, $I_x = 4A$ and $\hat{a}_\phi = +\hat{a}_z$ (By right-hand rule)

$$\therefore \vec{H}_x = \frac{4}{2\pi(1)}(\hat{a}_z)$$

Similarly, in y-direction:

$$\vec{H}_y = \frac{2}{2\pi(2)}(-\hat{a}_z)$$

$$\therefore \vec{H} = \frac{4}{2\pi(1)}(\hat{a}_z) + \frac{1}{2\pi(1)}(-\hat{a}_z)$$

$$\vec{H} = \frac{3}{2\pi}(\hat{a}_z) \text{ or } \frac{3}{2\pi}\hat{k} \text{ A/m}$$

The correct option is C).

Previous Years' Question

The flux density at a point in space is given by $\vec{B} = 4x\hat{a}_x + 2ky\hat{a}_y + 8\hat{a}_z$ Wb/m^2. The value of constant k must be equal to:

a) −2
b) −0.5
c) 0.5
d) 2

MAGNETIC SCALAR AND VECTOR POTENTIALS

- The electric field intensity $\vec{E}$ and the electric potential V are related as follows:

$$\vec{E} = -\nabla V$$

- Also, the magnetic potential can be scalar quantity (V_m) or vector quantity $\vec{A}$.
- To determine V_m and $\vec{A}$ involves the following two important identities.

Since, $\nabla \times (\nabla V) = 0$

$\therefore \nabla \cdot (\nabla \times \vec{A}) = 0$

which always hold valid for any scalar field V and vector field $\vec{A}$.

- Just as $\vec{E} = -\nabla V$, we can define the magnetic scalar potential V_m (in amperes) is related to $\vec{H}$ according to

  $\vec{H} = -\nabla V_m \qquad \{\text{if } \vec{J}=0\}$

  $\vec{J} = \nabla \times \vec{H} = \nabla \times (-\nabla V_m) = 0$

- The magnetic scalar potential V_m is only defined in a region where $\vec{J} = 0$ as in the above equation.
- Here, V_m follows Laplace's equation just as V does for electrostatic fields.
  Thus, $\nabla^2 V_m = 0 \qquad (\text{For } \vec{J}=0)$
  Since we know that for a magneto-static field, $\nabla \cdot \vec{B} = 0$.
- To satisfy the above equations (we can define the vector magnetic potential $\vec{A}$ (in Wb/m) such that:

  $\vec{B} = \nabla \times \vec{A} \qquad \ldots(4.44)$

  We have, $V = \int \frac{dQ}{4\pi \epsilon_0 R}$

- For line current: $\vec{A} = \int_L \frac{\mu_0 I d\vec{l}}{4\pi R}$
- For surface current: $\vec{A} = \int_S \frac{\mu_0 \vec{K} d\vec{S}}{4\pi R}$
- For volume current: $\vec{A} = \int_V \frac{\mu_0 \vec{J}\, dV}{4\pi R}$

Previous Years' Question

If a solid iron cylinder is placed in a uniform magnetic field, with its axis parallel to the magnetic field's direction. The magnetic field lines inside the cylinder will:

a) Vanishes from the inside of the cylinder.
b) Bend away from the axis.
c) Bend closer to the cylinder.
d) Keeps the same uniformity as before.

By substituting $\psi = \int_S \vec{B} \cdot d\vec{S}$ and employing Stokes's theorem, we obtain -

$$\psi = \iint_S \vec{B} \cdot d\vec{S} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S} = \oint_L \vec{A} \cdot d\vec{l}$$

FORCES DUE TO MAGNETIC FIELDS

- The force can be experienced due to magnetic fields by–
  - **i)** Due to a motion of the charged particle in a magnetic field,
  - **ii)** A current element in an external magnetic field.
  - **iii)** Between the two current-elements.

Force on a charged particle:

The electric force ($\vec{F}_e$) on a stationary-charge or electric charge (Q) moving in an electric field is expressed by Coulomb's experimental law and is linked to the electric field intensity ($\vec{E}$) as:

$$\vec{F}_e = Q\vec{E}$$

But, if Q is positive, then $\vec{F}_e$ and $\vec{E}$ will have the same directions.

- Also, the magnetic field can only exert force on a moving charge.
- The magnetic force ($\vec{F}_m$) on a charge (Q) moving with a velocity $\vec{v}$ in a magnetic field ($\vec{B}$) is:

$$\vec{F}_m = Q(\vec{v} \times \vec{B}) \quad ...(4.45)$$

Here, $\vec{F}_m$ is perpendicular to both $\vec{v}$ and $\vec{B}$.

- The total force on a moving charge (Q) in the presence of both fields is expressed by:

$$\vec{F} = \vec{F}_e + \vec{F}_m$$

$$\vec{F} = Q(\vec{E} + \vec{v} \times \vec{B}) \quad ...(4.46)$$

This is referred to as the Lorentz force equation.

Previous Years' Question

If magnetic field intensity at the region- 1 (μ_{r1})is given as $\vec{H}_1 = 7\hat{a}_x - 3\hat{a}_y + 6\hat{a}_z$. And at region-2($\mu_{r2}$) is $\vec{H}_2 = 15\hat{a}_x - 3\hat{a}_y + 12\hat{a}_z$.

The boundary is parallel to the coordinate axis.

a) the boundary doesn't have a surface current.

b) the boundary may have a surface current.

c) the boundary has $\vec{K}$ in $\hat{a}_x$ and $\hat{a}_y$ only.

d) the boundary has $\vec{K}$ in $\hat{a}_y$ and $\hat{a}_z$ only.

- Generally, it relates mechanical force to electrical force. If the charged a particle with mass 'm' is moving in a $\vec{E}$ and $\vec{B}$ fields, then due to Newton's second law of motion, we get:

$$\vec{F} = m\frac{d\vec{v}}{dt} = Q(\vec{E} + \vec{v} \times \vec{B}) \quad \ldots(4.47)$$

Force on a current-element:

The force exerted on a current-element $Id\vec{l}$ of a current-carrying conductor due to the magnetic field $\vec{B}$:

$$\vec{J} = \rho_V \vec{v}$$

- As we know that the relationship between current-elements is:

$$Id\vec{l} = \vec{K}dS = \vec{J}dV$$

or $Id\vec{l} = \rho_V \vec{v} dV = dQ\vec{v}$

Also, $Id\vec{l} = \frac{dQ}{dt}d\vec{l} = dQ\frac{d\vec{l}}{dt} = dQ\vec{v}$

Therefore, $Id\vec{l} = dQ\vec{v}$

- This demonstrates that an elemental-charge (dQ) moving with velocity $\vec{v}$ (correspondingly producing the convection current $dQ\vec{v}$) is similar to the conduction current-element $Id\vec{l}$.

  Thus, the force on a current-element $Id\vec{l}$ in a magnetic field $\vec{B}$ is found from equation (4.45) by replacing $Q\vec{v}$ by $Id\vec{l}$ that is, $d\vec{F} = Id\vec{l} \times \vec{B}$

- If the current (I) is flowing through a closed path l or in a circuit, then the force will be:

$$\vec{F} = \oint_L Id\vec{l} \times \vec{B} \quad \ldots..(4.48)$$

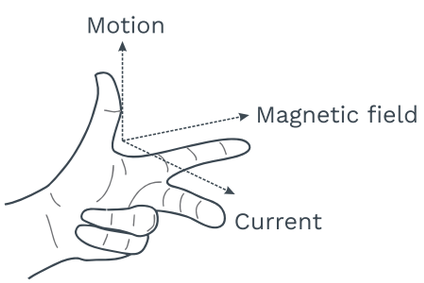


Fig. 4.36 Left-Hand Rule

Force between two current elements:

- Consider two elements $I_1 d\vec{l}_1$ and $I_2 d\vec{l}_2$. On applying Biot-Savart's law, both the two current elements will produce magnetic fields. Thus, the force $d(d\vec{F}_1)$ on the element $I_1 d\vec{l}_1$ due to the field $d\vec{B}_2$ produced by the element $I_2 d\vec{l}_2$.

  From the above equation, $d(d\vec{F}_1) = I_1 d\vec{l}_1 \times d\vec{B}_2$

- From Biot-Savart's law, we have: $d\vec{B}_2 = \frac{\mu_0 I_2 d\vec{l}_2 \times \hat{a}_{R_{21}}}{4\pi R_{21}^2}$

- Therefore, $d(d\vec{F}_1) = \frac{\mu_0 I_1 d\vec{l}_1 \times (I_2 d\vec{l}_2 \times \hat{a}_{R_{21}})}{4\pi R_{21}^2}$

  Or the total force $\vec{F}_1$ on current loop-1 due to current loop-2 is:

$$\vec{F}_1 = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{L_1} \oint_{L_2} \frac{d\vec{l}_1 \times (d\vec{l}_2 \times \hat{a}_{R_{21}})}{R_{21}^2} \quad ...(4.49)$$

MAGNETIC BOUNDARY CONDITIONS

Boundary conditions for normal components:

If we have two different mediums and the magnetic field in one region is known whereas the magnetic field in another region is unknown then, we apply the boundary conditions:

$\vec{B}_{n1} = \vec{B}_{n2}$ (Continuous) ...(4.50)

$\mu_1 \vec{H}_{n1} = \mu_2 \vec{H}_{n2}$

$\frac{\vec{H}_{n1}}{\vec{H}_{n2}} = \frac{\mu_2}{\mu_1} = \frac{\mu_0 \mu_{r2}}{\mu_0 \mu_{r1}}$

$\frac{\vec{H}_{n1}}{\vec{H}_{n2}} = \frac{\mu_2}{\mu_1} = \frac{\mu_{r2}}{\mu_{r1}}$ (Discontinuous) ...(4.51)

Previous Years' Question

The following four-vector fields are given in the cartesian coordinate system. The vector field which does not satisfy the property of magnetic flux density is:

a) $y^2\hat{a}_x + z^2\hat{a}_y + x^2\hat{a}_z$

b) $z^2\hat{a}_x + x^2\hat{a}_y + y^2\hat{a}_z$

c) $x^2\hat{a}_x + y^2\hat{a}_y + z^2\hat{a}_z$

d) $y^2z^2\hat{a}_x + x^2z^2\hat{a}_y + x^2y^2\hat{a}_z$

The normal component of $\vec{B}$ is continuous while that of $\vec{H}$ is discontinuous at the boundary.

Boundary condition for tangential components:

- On a plane surface, if the current is not flowing, then the tangential components will be equal.
  Then, mathematically:

$$\vec{H}_{t1} = \vec{H}_{t2} \qquad \text{... (4.52)}$$

$$\frac{\vec{B}_{t1}}{\mu_1} = \frac{\vec{B}_{t2}}{\mu_2}$$

$$\frac{\vec{B}_{t1}}{\vec{B}_{t2}} = \frac{\mu_1}{\mu_2} = \frac{\mu_{r1}\mu_0}{\mu_{r2}\mu_0}$$

$$\frac{\vec{B}_{t1}}{\vec{B}_{t2}} = \frac{\mu_1}{\mu_2} = \frac{\mu_{r1}}{\mu_{r2}}$$

- Here, at the boundary, the tangential components of $\vec{H}$ are continuous, whereas the tangential component of $\vec{B}$ is discontinuous.
- If there is a surface having 2 mediums and current flow is there on it, then the magnetic field will be as follows due to medium-1 & medium-2.

$$\left(\vec{H}_1 - \vec{H}_2\right)\hat{a}_{n12} = \vec{K}$$

$\vec{K}$ = Surface current density

$\vec{a}_{n12}$ = Unit vector normal to the interface and directed from medium-1 and medium-2.

$$\vec{H}_{t1} - \vec{H}_{t2} = \hat{a}_{n12} \times \vec{K}$$

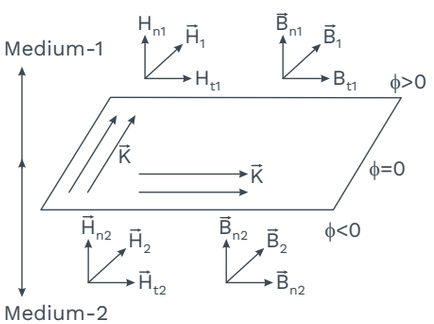


Fig. 4.37 Boundary Conditions

- Incidence and Refraction of magnetic fields:

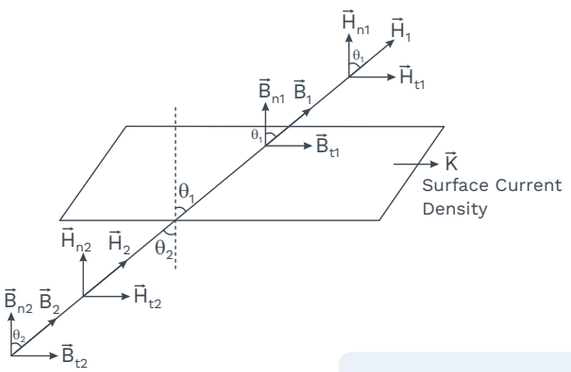


Fig. 4.38 Boundary Conditions

From the above figure, we get:

$$\tan\theta_1 = \frac{|\vec{B}_{t1}|}{|\vec{B}_{n1}|} = \frac{|\vec{H}_{t1}|}{|\vec{H}_{n1}|}$$

$$\text{and } \tan\theta_2 = \frac{|\vec{H}_{t2}|}{|\vec{H}_{n2}|} = \frac{|\vec{B}_{t2}|}{|\vec{B}_{n2}|} \quad \text{....(4.53)}$$

Boundary condition:

- The magnetic flux density normal component:

$$\vec{B}_{n1} = \vec{B}_{n2}$$

$$\vec{B}_1 \cos\theta_1 = \vec{B}_2 \cos\theta_2 \quad \text{... (4.54)}$$

$$\text{and } \vec{H}_{t1} - \vec{H}_{t2} = \hat{a}_{n12} \times \vec{K}$$

$$\vec{H}_1 \sin\theta_1 - \vec{H}_2 \sin\theta_2 = \hat{a}_{n12} \times \vec{K} \quad \text{... (4.55)}$$

If $\vec{K} = 0$

Equation (4.55) will become -

$$\vec{H}_1 \sin\theta_1 = \vec{H}_2 \sin\theta_2 \quad \text{... (4.56)}$$

Now, divide equation (4.56) by equation (4.54), we get:

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_{r1}}{\mu_{r2}} \quad \text{... (4.57)}$$

The equation (4.57) is valid only if $\vec{K} = 0$.

Previous Years' Question

A charged particle begins to move from the origin in a region that has a uniform magnetic field in the x-direction and a uniform electric field in the y-direction. its speed v when it reaches the point (x, y, z). v will depend on:

a) x-component.
b) y-component.
c) on both x and y-components but not on z-component.
d) on x, y, and z components.

Magnetic field at the centre of a regular polygon:

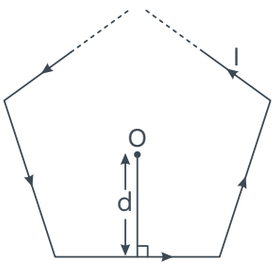
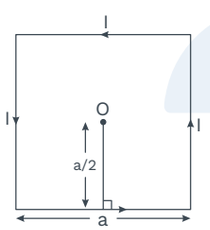


Fig. 4.39 Regular Polygon

Consider the regular polygon having n sides as shown in the above figure:

Then, $B_0 = \frac{\mu}{2\pi}\frac{I}{d} n \sin\left(\frac{\pi}{n}\right)$ Tesla ...(4.58)

Example 9:
Calculate the magnetic field density for the square as shown in the figure below:



Sol:
Here sides, n = 4
The magnetic field density for the square will be:

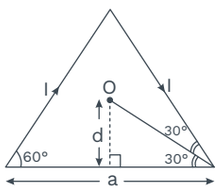
$$\therefore \quad B_0 = \frac{\mu_0}{2\pi}\frac{I}{(a/2)} 4 \sin\left(\frac{\pi}{4}\right)$$

$$B_0 = \frac{\mu_0}{\pi}\frac{I}{a} \times \frac{4}{\sqrt{2}}$$

$$B_0 = \frac{\mu_0}{\pi}\frac{I}{a} \times 2\sqrt{2} \quad \text{Tesla}$$

Example 10:
Calculate the magnetic field density for the equilateral triangle.

Sol:
Here, number of sides, n = 3



The magnetic field density for the equilateral triangle:

$$B = \frac{\mu}{2\pi}\frac{I}{d} n \sin\left(\frac{\pi}{n}\right)$$

here, $\tan 30° = \frac{d}{a/2}$

$$d = \frac{a}{2\sqrt{3}}$$

Also, n = 3

$$\therefore \quad B = \frac{9I}{2\pi a} \text{ Tesla}$$

Example 11:
Consider the magnetic vector potential given as follows:

$$\vec{A} = x^2 y\hat{a}_x + y^2 x\hat{a}_y - 4xyz\hat{a}_z$$

i) Calculate magnetic flux density at (–1, 2, 5)
ii) Calculate magnetic flux passing through surface z = 1; $0 \le x \le 1$; $-1 \le y \le 4$

Sol:

i) $$\vec{B} = \nabla \times \vec{A} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y & y^2 x & -4xyz \end{vmatrix}$$

$$\vec{B} = -4xz\hat{a}_x + 4yz\hat{a}_y + (y^2 - x^2)\hat{a}_z \quad (\text{Wb/m}^2)$$

$$\vec{B} = -4xz\hat{a}_x + 4yz\hat{a}_y + (y^2 - x^2)\hat{a}_z$$

The magnetic flux density is $\vec{B}\big|_{at(-1,\,2,\,5)} = 20\,\hat{a}_x + 40\,\hat{a}_y + 3\,\hat{a}_z (Wb/m^2)$

ii) The magnetic flux passing through surface $z = 1;\ 0 \le x \le 1;\ -1 \le y \le 4$ will be -

$$\psi_m = \int\int_s \vec{B} \cdot d\vec{S}$$

$$\psi_m = \int_{x=0}^{1} \int_{y=-1}^{4} (y^2 - x^2)\,dx\,dy = 20\,Wb$$

$(\because\ d\vec{S} = dx\,dy\,\hat{a}_z)$

MAGNETIC TORQUE AND MOMENT

- The torque $\vec{T}$ or mechanical moment of force on the loop is defined as the vector product of moment arm $\vec{r}$ and the force $\vec{F}$.

$$\vec{T} = \vec{r} \times \vec{F} \quad (Nm) \qquad ...(4.59)$$

- Considering the force on a current-loop in the magnetic field, the torque can be determined.
- If a loop is located parallel to the magnetic field, then it will experience a force that will tend to rotate it.

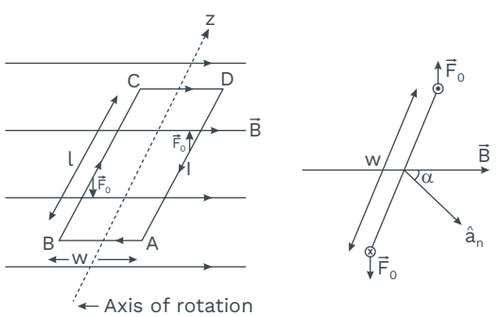


Fig. 4.40 Rectangular Loop

- Considering a rectangular loop with length 'l' and width 'w', which is placed in a uniform magnetic field $\vec{B}$ (as shown in the figure above).
- From this figure, the dl is parallel to $\vec{B}$ along with sides AB and CD of the loop thus, there will be no force exerted on those sides.

MAGNETOMOTIVE FORCE

- Magnetic devices like the toroid, transformers, motors and generators can be considered magnetic circuits.
- The magnetomotive force $\left(\vec{F}_m\right)$ is described as:

$$\vec{F}_m = NI = \oint \vec{H} \cdot d\vec{l}$$

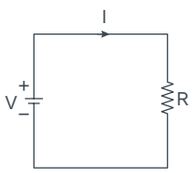


Fig. 4.41 a) An Electric Circuit

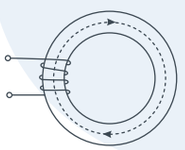


Fig. 4.41 b) A Magnetic Circuit

The source of magnetomotive force in magnetic circuits is generally a coil carrying current.

- The reluctance R, in Ampere-turns per weber as:

$$R = \frac{l}{\mu S} \qquad \ldots(4.60)$$

where l = Mean length
S = Cross-sectional area of the magnetic core.

- The permeance (P) is defined as the reciprocal of reluctance (R).
According to ohm's law: V = IR
The basic relationship for circuit elements will be:

$$F_m = \psi R$$

where ψ = Magnetic flux
R = Reluctance

- Like in the electric circuit, based on the above equation, the KCL and KVL laws can be applied to the nodes and the loops of a given magnetic circuit.

Similarly,
The rules of adding voltage* and combining the series and parallel resistances also apply for MMFs and reluctance.
∴ For n-magnetic circuit elements arranged in series:

$$\psi_1 = \psi_2 = \psi_3 = \ldots\ldots = \psi_n$$

and $F_m = (F_m)_1 + (F_m)_2 + (F_m)_3 + \ldots\ldots (F_m)_n$

- For n-magnetic circuit elements arranged in parallel connections:

$$\psi = \psi_1 + \psi_2 + \psi_3 + \ldots\ldots + \psi_n$$

and $(F_m)_1 = (F_m)_2 = (F_m)_3 \ldots = (F_m)_n$ ...(4.61)

Example 12:
If a 6.66A current enters a right-circular cylinder having a radius 5 cm. Then, the linear surface current density (A/m) at the end surface will be:

a) $\frac{66.6}{\pi}$(mA/m) **b)** $\frac{6.66}{\pi}$ (A/m) **c)** $\frac{66.6}{2\pi}$(A/m) **d)** $\frac{66.6}{\pi}$(A/m)

Sol:
Given,
Current, I = 6.66 A
The radius of the cylinder = 5 cm = 5×10^{-2} m.
The surface current density K is defined as-

$$K = \frac{I}{\text{Cross} - \text{sectional length}} \text{ (A/m)}$$

$$K = \frac{I}{2\pi r} = \frac{6.66}{2\pi(5 \times 10^{-2})} = \frac{6.66 \times 10^2}{10\pi} = \frac{66.6}{\pi}\text{(A/m)}$$

Hence, the correct answer is D).

Example 13:
Consider a region $z \leq 0$ which is a perfect conductor. The surface current density $\vec{K}$ is $4\hat{a}_x - 8\hat{a}_y$ (A/m) on its surface at the origin. If the region $z > 0$ is a free space, then the magnetic field intensity $\vec{H}$ in A/m, at the origin would be:

a) 0 **b)** $4\hat{a}_x - 8\hat{a}_y$

c) $8\hat{a}_x + 4\hat{a}_y$ **d)** $8\hat{a}_x - 4\hat{a}_y$

Sol:

Given that:

The surface current density on the surface of the region at origin is:

$$\vec{K} = 4\hat{a}_x - 8\hat{a}_y \text{ (A/m)}$$

And the magnetic field intensity $\vec{H}$ is given as:

$$\vec{H} = \vec{K} \times \hat{a}_n$$

$$\vec{H} = (4\hat{a}_x - 8\hat{a}_y) \times (-\hat{a}_z) \qquad (\because \hat{a}_n = -\hat{a}_z)$$

$$\vec{H} = 4\hat{a}_y + 8\hat{a}_x$$

$$\vec{H} = 8\hat{a}_x + 4\hat{a}_y$$

Hence, the correct option is C).

Chapter Summary



- Biot-Savart's law is identical to the Coulomb's law, and it states that the magnetic field intensity $d\vec{H}$ at $\vec{r}$ due to the current element $Id\vec{l}$ is:

  $$d\vec{H} = \frac{Id\vec{l} \times R}{2\pi R^3} \text{ (in A/m) where } R = r - \vec{r}' \text{ and } \vec{R} = |\vec{R}|.$$

- For a surface or volume current distribution, we replace $Id\vec{l}$ with $\vec{K}d\vec{S}$ or $\vec{J}dV$, respectively such, that: $Id\vec{l} \equiv \vec{K}d\vec{S} \equiv \vec{J}dV$.
- Ampere's circuit law is similar to the Gauss's law, and it states that the line integral of $\vec{H}$ around a closed path is equivalent to the total current enclosed by the path:

  $$\oint_L \vec{H} \cdot d\vec{l} = I_{enc} = \int_S \vec{J} \cdot d\vec{S}$$

  or $\nabla \times \vec{H} = \vec{J}$

- The magnetic flux through a surface S is given by:
  $\psi = \int_S \vec{B} \cdot d\vec{S}$ where $\vec{B}$ is the magnetic flux density (in Wb/m^2).
- In free space $\vec{B} = \mu_0 \vec{H}$ where the permeability of free space is $\mu_0 = 4\pi \times 10^{-7}$ H/m.
- The magnetic scalar potential V_m is defined as:

  $$\vec{H} = -\nabla V_m \quad \text{if } \vec{J} = 0.$$

- For a current-element $Id\vec{l}$ at $\vec{r}'$, the magnetic vector potential at $\vec{r}$ is:

  $$\vec{A} = \int \frac{\mu_0 Id\vec{l}}{4\pi R} \quad \text{if } R = |\vec{r} - \vec{r}'|$$

PRACTICE QUESTIONS

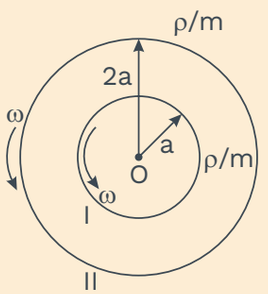
Q1 **Consider the two parallel wires separated by a distance 'D' and carry current 'I' in the same direction. The magnetic field along a line parallel to these wires and midway between them will be:**

a) depends upon 'D' only.
b) depends upon 'I' only.
c) zero.
d) depends on the permeability of the medium between the wires only.

Q2 **The magnetic field, due to an infinite linear current-carrying conductor, is:**

a) $H = \frac{mI}{2pr}$ A/m **b)** $H = \frac{mI}{4pr}$ A/m **c)** $H = \frac{I}{r}$ A/m **d)** $H = \frac{I}{2pr}$ A/m

Q3 **Consider the two concentric rings carrying equal uniform charge densities while revolving at the same angular speed (ω) about their common axis, passing through O as shown. The ratio of flux densities due to the two rings at O will be:**



a) 1 : 1 **b) 4 : 1** **c) 2 : 5** **d) 4 : 9**

Q4 **Consider an infinite number of concentric circular loops, each carrying current 'I' but alternately in opposite directions having radii of the loops as R, 2R, 4R ..... so on, in geometric progression. The magnetic flux density at the centre of the loops will be:**

a) $\frac{m_0 I}{3R}$ **b)** $\frac{m_0 I}{6R}$ **c)** $\frac{m_0 I}{4R}$ **d) Zero**

Q5 Current density $(\vec{J})$ in cylindrical coordinate system is given as:

$J(\rho, \theta, z) = 0$ for $0 < \rho < a$

$$J(r, \theta, z) = J_0 \frac{a}{\rho} \hat{i}_z \text{ for } a < \rho < b$$

where $\hat{i}_z$ is the unit vector along the z-coordinate axis. In the region, $a < \rho < b$, what is the expression for the magnitude of magnetic field intensity $\vec{H}$?

a) $\frac{J_0}{r^2}(r^3 - a^3)$

b) $\frac{J_0(r^3 - a^3)}{3a^2 r}$

c) $\frac{J_0}{r^2}(r^3 + a^3)$

d) $\frac{J_0}{4\pi r}(r^3 - a^3)$

Q6 What is the magnetic field intensity vector $\vec{H}$ between two parallel sheets with separation 'd' along the z-axis, both sheets carrying surface current $\vec{k} = k_y \hat{a}_y$?

a) $-k_y \hat{a}_y$ b) $-k_y \hat{a}_x$ c) $k_y \hat{a}_y$ d) Zero

Q7 Which of the following statement(s) is/are correct for Lorentz force?

a) The acceleration is normal to the plane containing the path of the particle and $\vec{B}$.

b) If the direction of the path of the particle is normal to $\vec{B}$, the acceleration will be maximum.

c) If the particle is at rest, then the field will deflect the particle.

d) If the path of the particle is in the same direction of $\vec{B}$, there will be no acceleration.

Q8 If a uniform and constant magnetic field exist in the z-direction in a vacuum. And a particle of mass 'm' with the same charge 'q' is introduced into this region with an initial velocity of $\vec{v} = v_x \hat{a}_x + v_z \hat{a}_z$. Assume that $\vec{B}$, m, q, v_x, and v_y are all non-zero. The particle will follow:

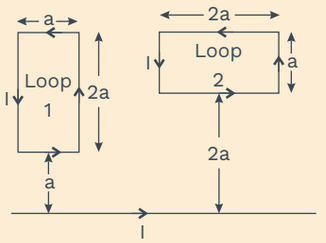
a) helical motion in the z-direction.

b) linear motion in the z-direction.

c) linear motion in the x-direction.

d) circular motion in the x-y plane.

Q9 **Consider the two rectangular loops and an infinite wire. All elements that carry current 'I' are shown in the given figure. If the force on loop-1 due to the infinite wire is 'F', then the force on the other loop due to the infinite wire is:**



a) $\frac{F}{2}$ b) F c) $\frac{F}{3}$ d) $\frac{3}{2}F$

Q10 **The force on the current-carrying conductor in a magnetic field depends on:**

a) Flux density
b) Current strength
c) Inductance
d) Direction of field and current

Answer Key:

Q1	c)	Q2	d)	Q3	a)	Q4	a)	Q5	a)
Q6	d)	Q7	a, b, d)	Q8	a)	Q9	a)	Q10	a, b, d)